

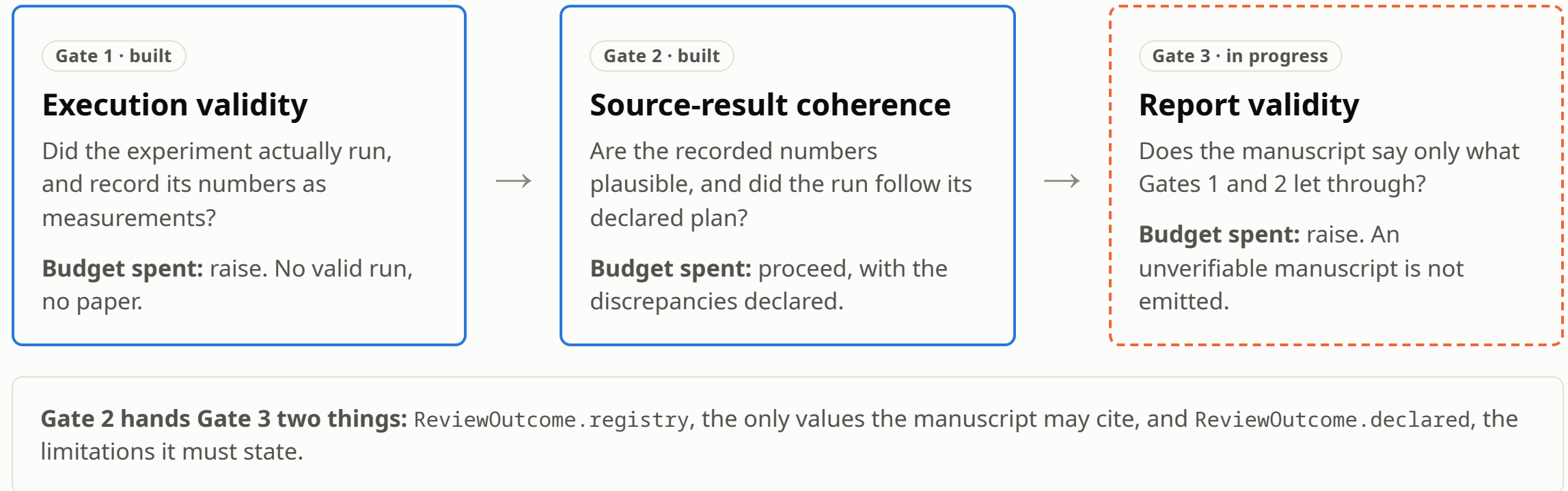
Report validity

Does every number and every citation in the manuscript trace to something the run actually measured or retrieved?

A flat list of deterministic checks. No model decides a verdict. A spent budget raises, and no manuscript is emitted.

Where Gate 3 sits

Three gates, three phases. The exhaustion policy differs on purpose.



Numbers are bound, not typed

The writer emits

```
\section{Results}  
Accuracy reaches \result{exp1.acc_at_400}  
with the full label budget.
```

The renderer writes

```
\section{Results}  
Accuracy reaches 0.97  
with the full label budget.
```

The renderer writes `str(value)` and nothing else. No rounding, so byte-identity with the registry is checkable.

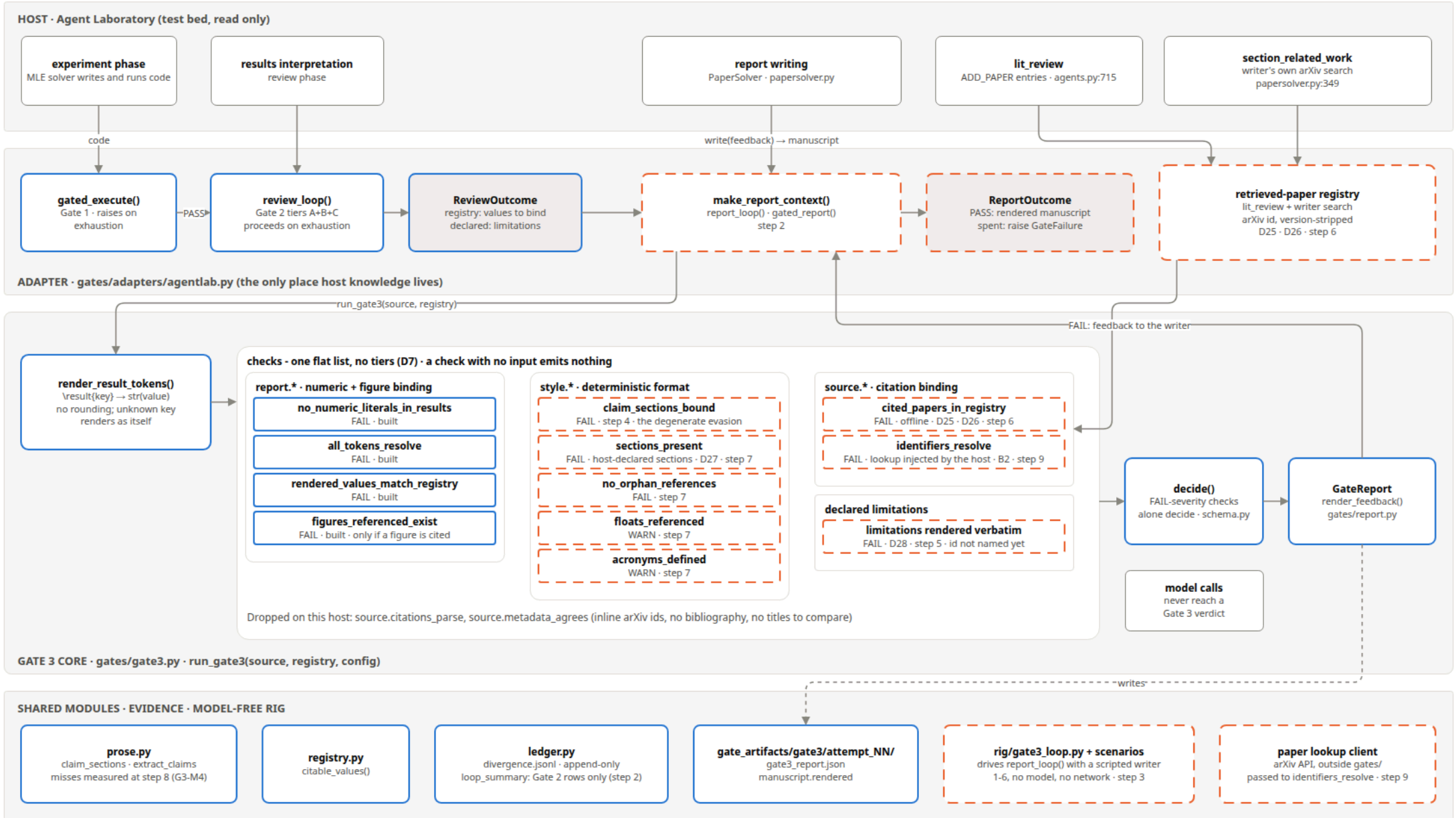
An unknown key renders as itself, so `report.all_tokens_resolve` fails instead of a sentence silently losing its number.

A numeral typed into a findings section fails
`report.no_numeric_literals_in_results`.

Limit, stated: that check is a scanner, and it misses some numbers ("87 percent", lines with `\cite`). Step 8 measures the miss rate.

G.A.T.E.S. Gate 3 - report validity stack

As of 4508891 · 2026-09-16 · solid blue = built · dashed orange = to build (step N) · grey = host, not ours · Dnn = decision in progress.md



The check list

CHECK	SEVERITY	STATUS	BASIS
report.no_numeric_literals_in_results	FAIL	 built	
report.all_tokens_resolve	FAIL	 built	
report.rendered_values_match_registry	FAIL	 built	
report.figures_referenced_exist	FAIL	 built	only if a figure is referenced
style.claim_sections_bound	FAIL	 step 4	results section that says nothing measurable
declared limitations present	FAIL	 step 5	D28 · id not named yet
source.cited_papers_in_registry	FAIL	 step 6	D25 · D26 · offline
style.sections_present	FAIL	 step 7	D27 · host-declared sections
style.no_orphan_references	FAIL	 step 7	
style.floats_referenced · style.acronyms_defined	WARN	 step 7	
source.identifiers_resolve	FAIL	 step 9	B2 · lookup injected by the host

Dropped on Agent Laboratory: source.citations_parse and source.metadata_agrees. The host cites inline arXiv ids with no bibliography and no titles, so both would have no input.

What reading the code changed

29

typed numbers in the archived gated manuscript

Across abstract, results and discussion. The old fixture read the readme and found 8. B4, closed in 4508891.

1 of 8

cited arXiv ids were in `lit_review`

The other 7 came from the writer's own per-section arXiv search (`papersolver.py:349`). A registry of `lit_review` alone would call them unretrieved.

0

bibliography entries

The host cites inline as (`arXiv 2410.21676v4`). Parsing a bibliography or comparing titles has nothing to read.

And one plan correction: `rig/loop.py` is wired to Gate 1 (`run_loop`, line 198), so it cannot be reused unchanged. Gate 3's loop follows Gate 2's as-built shape: a loop in the adapter, driven by a model-free rig.

Decisions taken

D25

What counts as retrieved

lit_review plus the writer's per-section arXiv search. The fabrication targeted is an id nothing retrieved. Amends D21.

D26

Canonical identifier

The arXiv id, compared version-stripped. A version mismatch is evidence, not a failure. No DOI fallback yet.

D27

Whose section standard

The host declares required sections at wiring time. No built-in list in gates/.

D28

Declared limitations

Rendered, not authored. The renderer inserts Gate 2's block verbatim, and Gate 3 checks it is there. A paraphrase cannot be matched deterministically.

Build order

1	B4: measure the manuscript, not the readme	 done	6	Paper registry + cited_papers_in_registry + scenario 4	 to build
2	Host entry point: report_loop(), gated_report()	 in progress	7	Remaining style.*, host-declared inputs only	 to build
3	Model-free rig, scenarios 1, 2, 3, 6	 to build	8	Measure scanner misses (G3-M4)	 to build
4	style.claim_sections_bound + scenario 5	 to build	9	source.identifiers_resolve, network last	 to build
5	Declared-limitations check (D28)	 to build			

Steps 1-8 are offline. If the network work stalls, Gate 3 is still complete, with a working loop and the fabricated-citation check.